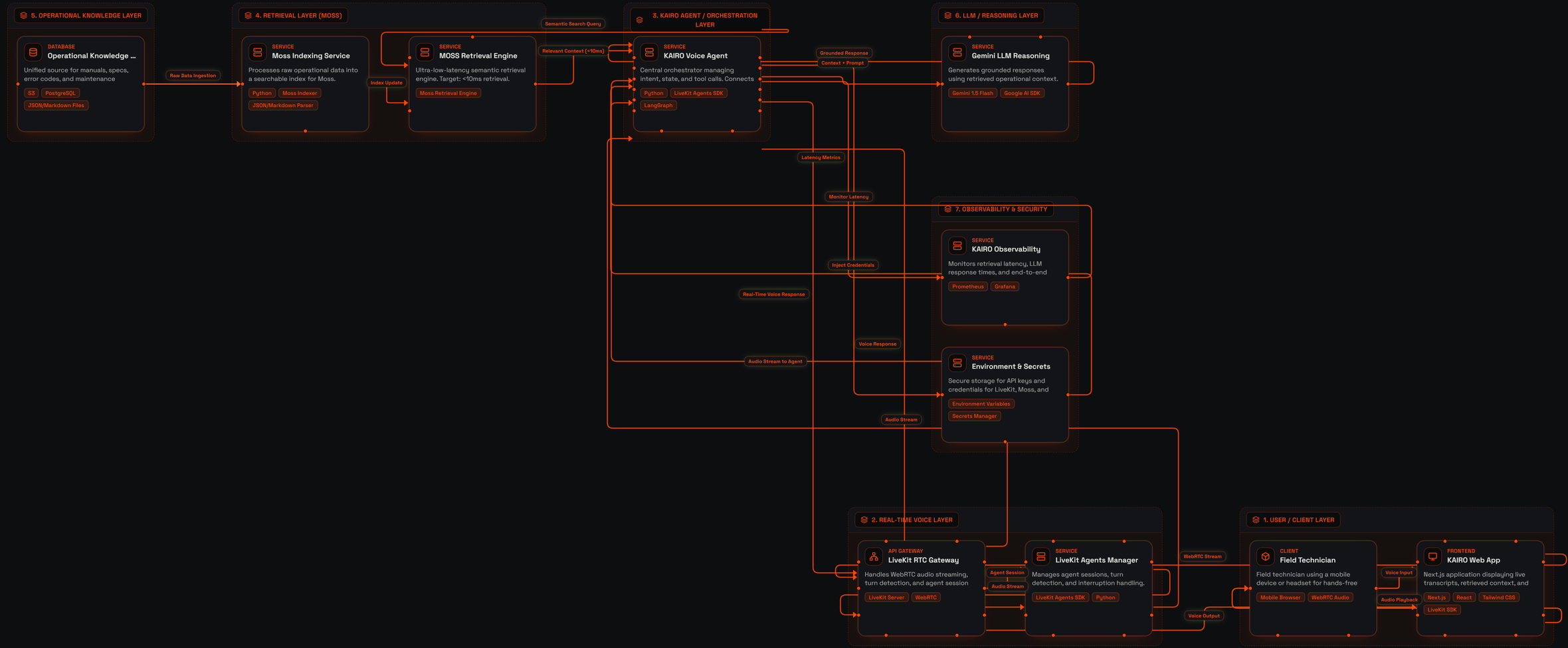
5. OPERATIONAL KNOWLEDGE LAYER
DATABASE
Operational Knowledge ...
Unified source for manuals, specs, error codes, and maintenance
S3
PostgreSQL
JSON/Markdown Files
Raw Data Ingestion
4. RETRIEVAL LAYER (MOSS)
SERVICE
Moss Indexing Service
Processes raw operational data into a searchable index for Moss.
Python
Moss Indexer
JSON/Markdown Parser
Index Update
SERVICE
MOSS Retrieval Engine
Ultra-low-latency semantic retrieval engine. Target: <10ms retrieval.
Moss Retrieval Engine
Semantic Search Query
Relevant Context (<10ms)
3. KAIRO AGENT / ORCHESTRATION LAYER
SERVICE
KAIRO Voice Agent
Central orchestrator managing intent, state, and tool calls. Connects
Python
LiveKit Agents SDK
LangGraph
Grounded Response
Context + Prompt
6. LLM / REASONING LAYER
SERVICE
Gemini LLM Reasoning
Generates grounded responses using retrieved operational context.
Gemini 1.5 Flash
Google AI SDK
Latency Metrics
Monitor Latency
7. OBSERVABILITY & SECURITY
SERVICE
KAIRO Observability
Monitors retrieval latency, LLM response times, and end-to-end
Prometheus
Grafana
Inject Credentials
Real-Time Voice Response
Voice Response
Audio Stream to Agent
SERVICE
Environment & Secrets
Secure storage for API keys and credentials for LiveKit, Moss, and
Environment Variables
Secrets Manager
Audio Stream
2. REAL-TIME VOICE LAYER
API GATEWAY
LiveKit RTC Gateway
Handles WebRTC audio streaming, turn detection, and agent session
LiveKit Server
WebRTC
Agent Session
Audio Stream
SERVICE
LiveKit Agents Manager
Manages agent sessions, turn detection, and interruption handling.
LiveKit Agents SDK
Python
WebRTC Stream
Voice Output
1. USER / CLIENT LAYER
CLIENT
Field Technician
Field technician using a mobile device or headset for hands-free
Mobile Browser
WebRTC Audio
Voice Input
Audio Playback
FRONTEND
KAIRO Web App
Next.js application displaying live transcripts, retrieved context, and
Next.js
React
Tailwind CSS
LiveKit SDK